

// PROJECT PROPOSAL // GROUP 45



WOMEN TECHSTERS BOOTCAMP BY TECH4DEV • COHORT 5.0
// GROUP 45 - CYBER SHIELD — CAPSTONE PROJECT — 2026

CYBER SHIELD

HUMAN FIREWALL CHALLENGE

AI Scam Simulator + Immunity Trainer

PROJECT TYPE

Capstone Project

TRACK / SECTOR

Security and Safety

SDG ALIGNMENT

SDG 4 // 9 // 16

GROUP NAME

Group 45 - Cyber Shield

PROJECT ONE-LINER

"Cyber Shield doesn't just detect scams; it trains users to resist them."

CYBER SHIELD

// 01 PROBLEM STATEMENT

The Problem We Are Solving

Online scams are becoming increasingly sophisticated, targeting individuals through phishing emails, fake websites, social engineering messages, and fraudulent links.

While many cybersecurity tools focus on detecting scams, the human factor remains the weakest link. Employees, students, children, and elderly individuals often fall victim due to lack of practical awareness and real-world exposure.

Organizations also face major financial and reputational losses due to human error in handling cyber threats.

There is a need for an interactive and practical training solution that helps users recognize and resist scams before real attackers exploit them.

// 03 PROPOSED SOLUTION

What Is Cyber Shield?

Cyber Shield is an AI-powered interactive scam simulation platform designed to train users to identify and resist cyber threats.

Rather than simply detecting scams, the platform builds user immunity through realistic, hands-on experience.

KEY FEATURES

- AI-generated phishing emails and scam messages
- Fake login pages to test credential safety awareness
- Interactive escape-room style challenges
- Real-time feedback on user decisions
- Personalized cybersecurity improvement tips

Users learn by doing, making cybersecurity training engaging, practical, and memorable.

Roles and Responsibilities

A. TECHNICAL PROJECT MANAGER

(Best fit for Technical PM students)

RESPONSIBILITIES

- Plan project scope and objectives
- Create timeline and sprint plans
- Assign responsibilities and manage workflow
- Coordinate communication between technical teams
- Track risks and ensure timely delivery
- Prepare project documentation and reports

DELIVERABLES

- // Project roadmap
- // Task schedule
- // Risk management plan
- // Progress tracking dashboard

B. CYBERSECURITY ANALYST

(Cybersecurity students)

RESPONSIBILITIES

- Research modern cyber threats and scam patterns
- Design realistic phishing and scam scenarios
- Develop cybersecurity awareness content
- Identify user vulnerabilities
- Ensure platform follows security best practices

DELIVERABLES

- // Scam scenario database
- // Threat analysis report
- // User safety guidelines

C. SECURITY OPERATIONS AND RISK LEAD

(Cybersecurity students)

RESPONSIBILITIES

- Design the Human Risk Score framework
- Analyze how users respond to threats
- Map behavioral weaknesses
- Develop mitigation strategies
- Ensure ethical handling of user data

DELIVERABLES

- // Risk scoring model
- // Behavior analysis framework
- // Security compliance checklist

// 04 TEAM TASK SHARING (CONTINUED)

D. AI SYSTEMS COORDINATOR

(Cybersecurity student with coding OR Technical PM with tech skills)

RESPONSIBILITIES

- Plan AI feature implementation
- Coordinate between AI development and platform team
- Oversee scam scenario automation
- Manage AI workflow and testing
- Ensure AI outputs align with project goals

DELIVERABLES

- // AI feature plan
- // AI workflow diagrams
- // System integration documentation

E. TECHNICAL IMPLEMENTATION LEAD

(Cybersecurity student with coding background)

RESPONSIBILITIES

- Develop interactive simulation platform
- Build phishing email and fake login interfaces
- Integrate AI and scoring systems
- Manage backend functionality
- Ensure system reliability

DELIVERABLES

- // Functional application
- // Simulation engine
- // System integration

F. DOCUMENTATION AND PRESENTATION LEAD

(Technical PM students)

RESPONSIBILITIES

- Prepare project proposal and documentation
- Develop demo presentation slides
- Translate technical features into simple explanations
- Coordinate demo storytelling
- Prepare user guides

DELIVERABLES

- // Project documentation
- // Presentation slides
- // Demo script

SHARED RESPONSIBILITIES

Brainstorming features // Testing simulations // Demo rehearsal // Feedback improvements

// 05 AI COMPONENT

Artificial Intelligence Functions

SCENARIO GENERATION

AI creates realistic phishing emails, scam messages, and fraudulent web pages.

BEHAVIORAL ANALYSIS

AI monitors how users interact with simulations; click patterns, hesitation time, information entered.

RISK SCORING SYSTEM

AI calculates a Human Risk Score indicating how vulnerable a user is to scams.

PERSONALIZED TRAINING

AI recommends tailored safety tips and learning modules based on user weaknesses.

// 06 TARGET AUDIENCE

Who Is This Built For?

COMPANIES

Staff cybersecurity awareness and phishing resistance training.

SCHOOLS

Digital safety education integrated into curricula.

FAMILIES

Child online protection and parental scam guidance.

ELDERLY

Accessible scam awareness for the most vulnerable.

// **SEE PAGE 7** Full go-to-market strategy, budget, team costing & milestones in the Project Management Scope.

How We Built This

This project was built using a combination of AI tools, combined research, and collaborative development, each contributing a unique strength to the final product.



Claude

PRIMARY DEVELOPMENT AI

Frontend development, game logic, Behavioural Analysis Engine, UI/UX design, documentation, and presentation structure. Primary AI used throughout the entire project build.



ChatGPT

CONTENT & IDEATION

Brainstorming scam scenarios, refining problem statements, drafting initial proposal content and game narrative ideas. Used for early ideation and content generation.



Google Gemini

RESEARCH & VALIDATION

Cross-referencing cybersecurity statistics, validating scam patterns, and supporting SDG alignment research across Africa. Used to verify facts and regional data.



Google

DATA & STATISTICS

Sourcing cybercrime statistics, real-world scam examples, and best practices in cybersecurity awareness training. Foundation for all data referenced in this project.

■ COMBINED AI + HUMAN INTELLIGENCE APPROACH

Multiple AI tools were used collaboratively alongside team research and human insight, ensuring accuracy, realism, and impact.

// RESEARCH METHODOLOGY

- **Claude:** Used for all code generation, UI/UX design, documentation, PDF creation, and spreadsheet building. The primary engine behind the entire technical build.
- **ChatGPT:** Used in early ideation phases to brainstorm scam scenarios, draft content, and refine the problem statement narrative.
- **Google Gemini:** Used to cross-reference cybersecurity statistics specific to Africa, validate scam patterns, and confirm SDG alignment.
- **Google Search:** Used as the primary research engine to source real-world cybercrime statistics, phishing examples, regional scam data, and best practices in cybersecurity awareness training.

// 06 PROJECT MANAGEMENT SCOPE

Go-To-Market Strategy & Project Management

The following page presents the full Project Management Scope for Cyber Shield, developed as a companion document to this proposal. It outlines the complete go-to-market strategy, team structure, budget breakdown, 18-month delivery timeline, marketing campaign, revenue model, risk register, and key success milestones required to bring this platform to market.

BUDGET & COSTING

\$895,720 total investment across 18 months covering 14 roles, tech, marketing & contingency

TEAM & TIMELINE

14 specialists across 5 delivery phases over 18 months; Cybersecurity, AI, Dev, Legal & marketing

REVENUE & IMPACT

B2C, B2B SaaS, Government & White-label API. Year 2 Annual Recurring Revenue target: \$1.2M.

// SEE OVERLEAF - PAGE 7

The full Project Management Scope infographic is presented on the next page.

// THIS SECTION COVERS:

- 01 Team Professionals & Costing**
14 roles with monthly rates, durations and total costs
- 02 Project Timeline**
5-phase 18-month delivery roadmap to market launch
- 03 Full Budget Breakdown**
Personnel, infrastructure, marketing and contingency
- 04 Marketing Campaign Strategy**
Pre-launch, launch and growth phases with channels
- 05 Revenue & GTM Model**
B2C, B2B SaaS, government and white-label streams
- 06 Risk Register**
5 key risks with impact ratings and mitigations
- 07 Key Milestones**
7 success checkpoints from Month 3 to break-even



CYBER SHIELD

HUMAN FIREWALL CHALLENGE · AI SCAM SIMULATOR + IMMUNITY TRAINER

// PROJECT MANAGEMENT SCOPE

WOMEN TECHSTERS BOOTCAMP · COHORT 5.0 · GROUP 45 · 2026

BUDGET

MARKETING

TIMELINE

TEAM & COSTING

\$127,500

TOTAL PROJECT BUDGET

18 MOS

GO-TO-MARKET TIMELINE

14

TEAM PROFESSIONALS

\$35K

MARKETING BUDGET

5 PHASES

DELIVERY MILESTONES

// 01 TEAM PROFESSIONALS & COSTING

#	ROLE	DISCIPLINE	RATE/MONTH	DURATION	TOTAL COST
01	Technical Project Manager Scope, sprints, risk, delivery	Project Management	\$4,500	18 mos	\$81,000
02	Lead Cybersecurity Analyst Threat design, scam scenarios, pen testing	Cybersecurity	\$5,200	12 mos	\$62,400
03	Security Operations / Risk Lead Human Risk Score, compliance, data ethics	Cybersecurity	\$4,800	12 mos	\$57,600
04	AI / ML Engineer Behavioural engine, Claude API integration	AI Engineering	\$6,500	14 mos	\$91,000
05	Full-Stack Developer (Lead) React frontend, backend, simulation engine	Software Dev	\$5,800	14 mos	\$81,200
06	React Native Developer iOS & Android app (Phase 5)	Mobile Dev	\$5,000	6 mos	\$30,000
07	UI/UX Designer Game UI, escape room interfaces, branding	Design	\$3,800	10 mos	\$38,000
08	Legal / Compliance Officer POPIA/GDPR, ethical use, privacy policy	Legal	\$4,200	4 mos	\$16,800
09	Scrum Master / Agile Coach Sprint facilitation, retrospectives, velocity	Project Management	\$3,500	18 mos	\$63,000
10	Cybersecurity Content Strategist Safety tips, training copy, scenario writing	Cybersecurity	\$2,800	10 mos	\$28,000
11	QA / Test Engineer Game testing, security testing, UAT	QA Engineering	\$3,200	8 mos	\$25,600
12	Digital Marketing Manager Campaigns, SEO, social, partnerships	Marketing	\$3,500	12 mos	\$42,000
13	DevOps / Cloud Engineer AWS/GCP, CI/CD, infrastructure, uptime	DevOps	\$4,500	12 mos	\$54,000
14	Business Development Lead B2B sales, school & corp partnerships	Business Dev	\$4,000	12 mos	\$48,000
TOTAL PERSONNEL (18 MOS)			\$718,600		

// 02 PROJECT TIMELINE – 18 MONTHS TO MARKET

- PHASE 1 // MONTHS 01–03 · DISCOVERY & SCOPING
Requirements gathering, threat research, tech stack finalised, team onboarding, legal setup (POPIA/GDPR), branding locked.
- PHASE 2 // MONTHS 04–07 · CORE BUILD
5-level escape room developed, Behavioural Analysis Engine built, Claude API integrated, scam scenario database created, HUD + leaderboard launched.
- PHASE 3 // MONTHS 08–11 · TESTING & BETA
UAT with pilot schools & SMEs, QA security testing, Human Risk Score calibration, multiplayer mode developed, performance optimisation.
- PHASE 4 // MONTHS 12–15 · MARKETING LAUNCH
Full marketing campaign live, school curriculum pilots, corporate B2B outreach, PR push, digital certificate system, influencer partnerships.
- PHASE 5 // MONTHS 16–18 · MOBILE & SCALE
React Native iOS & Android app launch, regional Africa expansion, enterprise licensing, API white-label product for corporates.

// 03 FULL BUDGET BREAKDOWN

PERSONNEL		TECHNOLOGY & INFRASTRUCTURE	
Team Salaries (18 mos)	\$718,600	Cloud / Hosting (AWS/GCP)	\$14,400
Freelancers / Contractors	\$18,000	AI API (Claude) Costs	\$8,400
		Software Licences / Tools	\$6,000
MARKETING & GROWTH		OPERATIONAL & CONTINGENCY	
Digital Campaigns (Meta/Google)	\$12,000	Legal / Compliance / IP	\$8,500
Influencer / PR / Events	\$8,000	Admin / Office / Comms	\$5,400
Content Creation / Video	\$7,500	Contingency Reserve (10%)	\$81,420
SEO / Community / Socials	\$7,500		
TOTAL PROJECT INVESTMENT (18 MOS)	\$895,720	BREAK-EVEN TARGET	MONTH 22

// 04 MARKETING CAMPAIGN STRATEGY

PRE-LAUNCH // MONTHS 10–11

Beta waitlist campaign, teaser content on LinkedIn/X/TikTok, cybersecurity influencer seeding, press kit to African tech media (TechCabal, IT Web SA). Target: 5,000 sign-ups before launch.

LAUNCH // MONTH 12

Product Hunt launch, live demo event (virtual + Cape Town/Lagos), paid Meta & Google campaigns, school partnership press release, challenge hashtag #HumanFirewall.

GROWTH // MONTHS 13–18

Corporate B2B outreach via LinkedIn Sales Navigator, school curriculum pilot grants, referral programme (earn badges for sign-ups), quarterly threat reports for PR, App Store / Play Store featured listing push.

META ADS

GOOGLE ADS

SEO/CONTENT

INFLUENCERS

PR / MEDIA

B2B OUTREACH

PRODUCT HUNT

COMMUNITY

// 05 REVENUE & GO-TO-MARKET MODEL

B2C FREE + PREMIUM

Free: 2 game levels. Premium: \$9.99/mo — all 5 levels, AI coaching, certificate, leaderboard. Schools: \$2/student/year (volume licensing).

B2B CORPORATE LICENSING

SaaS platform: \$499–\$2,999/mo per company tier (SME to Enterprise). Includes staff dashboards, HR risk reports, branded modules.

GOVERNMENT / NGO CONTRACTS

Grant-funded deployments via digital safety ministries, cybercrime bureaux, and UNICEF / AU digital inclusion programmes.

WHITE-LABEL / API

License Cyber Shield engine to banks, telcos and insurers for customer cyber-literacy compliance. \$15K–\$50K setup + rev share.

YR 2 REVENUE TARGET

\$1.2M ARR

// 06 RISK REGISTER

RISK	IMPACT	MITIGATION
AI API cost overrun	HIGH	Rate limiting + caching layer
Data privacy breach	CRITICAL	No real PII stored; encrypted sessions
Talent retention	MED	Equity options + flexible contracts
Competitor entry	MED	Africa-first moat + community lock-in
Scope creep	HIGH	Strict MoSCoW in each sprint

// 07 KEY MILESTONES & SUCCESS METRICS

MO 03	Tech stack + legal entity registered
MO 07	MVP live with 5 levels + AI engine
MO 11	500 beta users — NPS target ≥ 65
MO 12	Public launch — 10K users month 1
MO 15	50 corporate clients signed
MO 18	Mobile app live — 100K+ total users
MO 22	Break-even — ARR approaching \$1.2M